

SAGNIK PAL

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RESUME SUMMARY

Analytical and detail-oriented B.Tech student with hands-on experience in real-time automation systems at IFB Industries Ltd. Skilled in data analysis, dashboarding, and problem-solving to drive operational efficiency and decision-making. Strong communicator with a data-driven mindset, eager to optimize customer operations and deliver impactful business outcomes.

EDUCATIONAL QUALIFICATIONS

Qualification	School/College	Board/University	Year	% / CGPA
B.Tech in E&CS	KIIT	KIIT	Ongoing (Sem 7)	8.67 CGPA
Class XII (PCM)	DPS Ruby Park, Kolkata	CBSE	2023	88.8%
Class X	DPS Joka, South Kolkata	CBSE	2021	91.4%

INTERNSHIP

IFB Industries Ltd - IoT and Embedded Systems Intern - May 2025 to July 2025 (Location - Goa)

Engineered real-time automation and monitoring systems, improving efficiency and customer experience through smart kiosks. Developed sensor-based energy optimization solutions and built real-time dashboards using WebSockets, MQTT, and REST APIs for data-driven decision-making.

ITC Infotech – IT Support Service Trainee - May 2026 to November 2026 (Location - Kolkata)

Provided end-to-end IT support and contributed to both backend and frontend development, ensuring seamless system performance and user experience. Worked on system patching and security hardening to maintain compliance and protect enterprise infrastructure. Assisted in troubleshooting hardware, software, and network issues, improving system reliability and reducing downtime.

Contributed to the development and enhancement of the ITC Pravesh onboarding portal, building and optimizing features across the full stack using modern web technologies. Integrated APIs and improved workflow automation for employee onboarding processes. Additionally, supported real-time monitoring solutions and dashboards using technologies like WebSockets, MQTT, and REST APIs to enable efficient data-driven operations.

LIVE PROJECTS

Kiosk-Based Smart System (IFB Industries Ltd – May 2025 to July 2025)

- Developed a touchscreen-based kiosk using HTML, CSS, JavaScript, and Node-RED to control and demonstrate smart appliances in a Kiosk .

Occupancy-Based Automation System (IFB Industries Ltd – May 2025 to July 2025)

- Designed a smart workspace automation solution using motion and IR sensors to detect presence and adjust electrical systems accordingly.

Smart Air Monitoring System (IFB Industries Ltd – May 2025 to July 2025)

- Built an environmental dashboard using MQTT and WebSockets to visualize temperature, humidity, and gas sensor data in real time.

Remote Controlled AeroModel (IIT Kharagpur – Jan 2025 to Mar 2025)

- Participated in National Aeromodelling Competition; built a radio-controlled aircraft with stability algorithms and remote navigation controls.

Home Automation System (Aug 2023 to Oct 2023)

- Designed an Arduino-based home automation circuit responding to sensor input and basic voice commands

Non-Invasive Glucose Monitoring (April 2025 to Jan 2026)

- Developed an optical-based glucose monitoring system using NIR spectroscopy, leveraging Beer–Lambert’s Law to estimate glucose concentration non-invasively.

CERTIFICATIONS

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| • IFB Industries – IoT & Embedded Systems Internship Certification | Completed: 28 July 2025 |
| • Scaler Academy – Database Management Systems (DBMS) | Completed: April 2025 |
| • Startup Bharat – KIIT University | 1st Position – March 2025 |
| • Business Plan Competition – IIT Kharagpur | 13th Position – March 2025 |

SKILLS

- Python, C/C++, Java, HTML, CSS, JavaScript, Node.js, Node-RED, MQTT, WebSocket, REST API, ngrok, Vivado, TINA, Product Management, Design Thinking, Sensor Interfacing, Real-Time Dashboards